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Developing numerical tools to improve the diagnosis of peripheral artery disease

Luke Barratt, Al Benson, Marc Bailey, Toni Lassila, Yasina Somani

University of Leeds

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Presenter: Luke J Barratt (University of Leeds)

Peripheral artery disease (PAD) is an age-related condition present in >10% of adults over 65 years and is characterised by atherosclerotic narrowing of arteries classically affecting the legs [1]. This reduces oxygen delivery to skeletal muscle and leads to pain during walking (intermittent claudication, IC) [2]. Early diagnosis of PAD is key to reducing risk of cardiovascular events; a risk that is two-fold greater in PAD than in those with stable coronary artery disease [3]. The gold-standard diagnostic method for PAD is the ankle-brachial index (ABI; the ratio of leg to arm systolic blood pressure), sometimes supplemented with ultrasound, MRI or CT [4]. However, traditional diagnostic methods do not always detect PAD, particularly in patients with calcified arteries or atypical presentations often occurring in women. As a result, PAD frequently remains underdiagnosed and undertreated compared with other cardiovascular conditions [5]. To address the limitations of current diagnostic methods, we are developing numerical methods to improve risk prediction algorithms by assessing factors local to the occluded limbs. Three mathematical models are being developed that can assess: (1) blood flow in the large arteries using 1D Navier-Stokes for flow in compliant tubes [6], (2) microvascular flow using Poiseuille's Law in networks [7], (3) dilation of vascular beds mediated by shear forces within the microvasculature and metabolic activity in the muscle; and (4) lumped oxygenation of the skeletal muscle using a coupled set of equations representing oxygen transport in the blood and muscle fibres and cellular metabolism [8]. Ultrasound and NIRS data during transition to exercise will be used to validate the numerical model. By linking vascular anatomy and function, plaque severity, and muscle metabolism on oxygen dynamics, this work aims to help identify key causes of IC in PAD patients and guide targeted strategies for improved diagnosis and treatment.

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